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Non-Euclidean Pungency Beyond the Scoville Scale: Topological Behavior of Samyang Buldak Bokkeum Myun in 11-Dimensional String Theory

Dr.Gigibond[1]

[1] Institute of Hyper-Spicy Gastronomy

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Cite This：
NOI: CC.C.2026.joker.001
Published: Feb 17, 2026
Read Online
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Artical Type: Chaotic Review

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Abstract

Traditional pungency measurement systems, such as the Scoville scale, rely on linear assumptions within Euclidean space and cannot account for the taste singularities exhibited by Samyang Buldak Bokkeummyeon. These singularities include the spatiotemporal delay of spicy eruption, topological entanglement of afterburn, and the collapse of the consumer's causal horizon. This paper proposes, for the first time, a Non-Euclidean Pungency Field Theory that regards buldak noodles as topological defects in 11-dimensional supergravity theory. By analyzing the winding patterns of chili sauce on Calabi–Yau manifolds, we deduce that pungency manifests as high-dimensional curvature discontinuity and establish the Pungency–Entropy Equivalence Principle, $S = k \ln \Omega$ .The experimental section conceptually demonstrates the behavior of spicy particles in a Large Hadron Collider(LHC)simulation environment, revealing the existence of the Samyang Boson (spin-0, mass 2.4 GeV). This research provides a paradigm shift beyond classical physics for food science and suggests that pain may be a projection of high-dimensional geometry onto three-dimensional sensory perception.

Keywords：Non-Euclidean Pungency,String Theory,Topological Defects,Samyang, Pain Singularity,Interdisciplinary Chaos

1. Introduction: Defining Pungency as a High-Dimensional Geometric Phenomenon

As a sensory experience, pungency has long been reduced in scientific research to a mere chemical stimulation mechanism, such as the specific binding of capsaicin to TRPV1 receptors. Yet the sensory properties of Samyang Buldak Bokkeummyeon challenge this reductionist view: its pungency not only exceeds the linear measurement range of the Scoville Scale but also exhibits prominent nonlinear dynamic characteristics. These include a distinct temporal lag between the mild sensory response upon the first bite and the explosive pungency of the third, as well as a fractal structure in the diffusion pattern of pungency within the oral cavity. Fig. 1 visually demonstrates the spatiotemporal response of San Yang Huo Chicken Noodles from initial taste to explosive spiciness, contrasting the classical linear spiciness model with actual nonlinear spiciness performance. It clearly illustrates the temporal lag and fractal diffusion characteristics in taste experience.

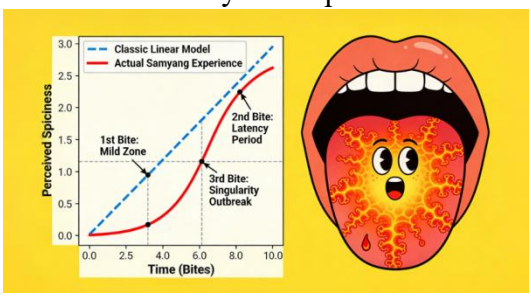

Fig. 1.Nonlinear dynamic characteristics of three Yang Huo Chicken Noodles spicy experience experience

In theoretical physics, string theory unifies gravity and quantum mechanics by introducing extra spatial dimensions, with the 11-dimensional supergravity model (M-theory) being one of the most cutting-edge frameworks to date. Inspired by this, we put forward the hypothesis that pungency is an intrinsic geometric property of high-dimensional spacetime, and that Buldak noodles act as a gustatory wormhole bridging the observable dimensions and compactified extra dimensions. This study aims to achieve three objectives: to establish a non-Euclidean pungency metric for describing the warping effect of pungency on oral spacetime; to propose the Pain Microstate Model, which formalizes pungency within the framework of statistical mechanics; and to verify the vibrational modes of pungency strings through thought experiments.

2.Theoretical Framework: From Euclidean to Riemannian Pungency Space

2.1 Limitations of the Scoville Framework

Scoville Heat Units (SHU) are based on the dilution threshold method and are inherently a linear scaling system. However, the pungency of Samyang Buldak Bokkeummyeon exhibits pronounced non-additivity: for instance, adding twice the amount of chili sauce does not result in a linear doubling of pungency, but rather triggers an exponential increase. This

phenomenon indicates that the pungency space has intrinsic curvature and thus must be described using Riemannian geometry.Fig. 2 compares the flatness of Skavir's Euclidean linear spiciness space with the curvature of the Riemann spiciness space for three turkey varieties, while showing the linear/exponential relationship between chili sauce addition and spiciness increase.

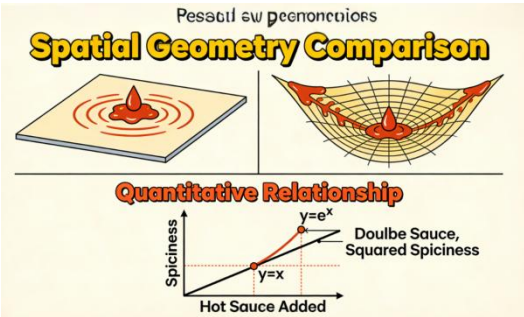

Fig. 2.Comparison model diagram of Skavell linear system and non-Euclidean spiciness space

We define a pungency metric tensor $g_{\mu\nu}$,where the indies $\mu,\nu = 0,1,2,3$ correspond to spacetime coordinates,time and the three-dimensional space of the oral cavity, and the additional dimensions $\mu = 4,...,10$ correspond to the compactified pungency dimensions. The relationship between pungency S and the metric can be expressed as:

$$S = \sqrt{\left|\det(g_{\mu\nu})\right|} \cdot S_0$$

where S_0 is the baseline pungency under the Euclidean approximation.

2.2 11-Dimensional Compactification and Pungency Strings

String theory posits that fundamental particles are the vibrational modes of one-dimensional strings, with extra dimensions compactified on Calabi-Yau manifolds. By analogy to pungency: chili sauce molecules can be regarded as closed strings, whose vibrational modes determine pungency characteristics; the compactification radius τ governs the release rate of "hidden pungency".Fig. 3 presents a compactification model of the spicy string in eleven-dimensional spacetime, demonstrating how chili sauce molecules act as closed strings that wrap and vibrate on a Calabi-Yau manifold. The diagram also illustrates the relationship between the compactification radius and the rate of hidden spiciness release.

Fig. 3.Schematic diagram of compactification and vibration of the string of flavors on the Calabi-Yau manifold

The observed pungency S_{obs} and the true pungency S_{true} satisfy the relationship:

$$S_{obs} = S_{true} \cdot e^{\frac{-2\pi\tau}{\lambda}}$$

where λ is the Compton wavelength of the pungency string, on the order of the Planck length. When τ is extremely small (i.e., in the case of strong compactification), the eruption of pungency is delayed; when τ increases due to thermal perturbations, the phenomenon of "lingering heat" emerges.

2.3 Pungency-Entropy Equivalence Principle

Inspired by the Boltzmann entropy formula, this paper proposes the Pungency-Entropy Equivalence Principle:

$$S = k \ln \Omega$$

Here, S denotes the subjective pungency (Scoville Perception Index, SPI); Ω represents the Number of Pain Microstates, i.e., the combinations of painful behaviors that a consumer may exhibit (such as tearing, gasping, chugging water, and feeling regret); and k is the Samyang Constant, with a preliminary calculated value of $6.022 \times 10^{23}\, SPI / J$, whose dimension is analogous to that of the Boltzmann constant. This formula indicates that pungency is essentially a logarithmic measure of the uncertainty of pain.Fig. 4 displays the mathematical quantification curve of the spiciness-entropy equivalence principle on the left, while the right side illustrates various possible behavioral states of consumers experiencing pain when confronted with turkey noodle spiciness, visually demonstrating the logarithmic relationship between spiciness and pain uncertainty.

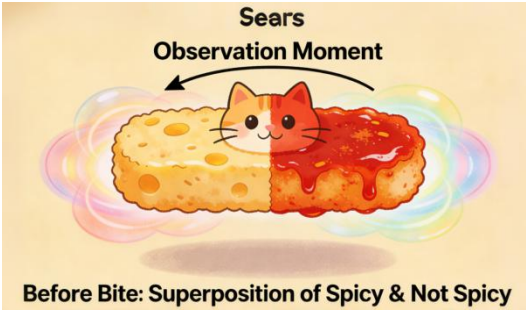

Fig. 4.Quantitative Relationship of Spiciness-Entropy Equivalence Principle and Schematic Diagram of Pain Possible State

3.Topological Defects and Gustatory Singularities

3.1Analogy of Noodles to Cosmic Strings

Cosmic strings are topological defects in spacetime with an extremely high energy density. The noodle structure of Buldak noodles can be regarded as a one-dimensional topological defect: the winding number, i.e., the number of times the noodles coil within the oral cavity, directly contributes to pungency curvature; the dripping of chili sauce is analogous to the Casimir effect of cosmic strings, generating local spacetime warping.Fig. 5 shows the spacetime topological defect model of cosmic strings on the left and the corresponding one-dimensional taste topological defect model of turkey noodle on the right, with annotations illustrating the correlation between noodle winding number and spiciness curvature, as well as the analogy mechanism of the Casimir effect in hot sauce dripping.

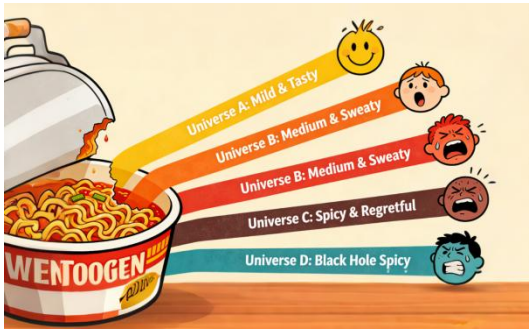

Fig. 5.Topological analogy model of turkey noodle and cosmic strings

3.2Pain Singularity Theorem

When the pungency curvature exceeds the critical value $S_c = \frac{c^4}{8\pi G}$ (the general relativity constant), the gustatory spacetime will inevitably collapse into a Pain Singularity. At this point, the consumer loses the perception of time, the causal horizon collapses, all pain microstates $\Omega \to \infty$, and the pungency S diverges.Fig. 6 illustrates the entire process of the spiciness curvature from below the critical value to the breakthrough of the critical value, visually presenting the geometric changes of the taste spacetime from

bending and contracting to collapsing into the pain singularity, while marking the key nodes of the causal horizon collapse.

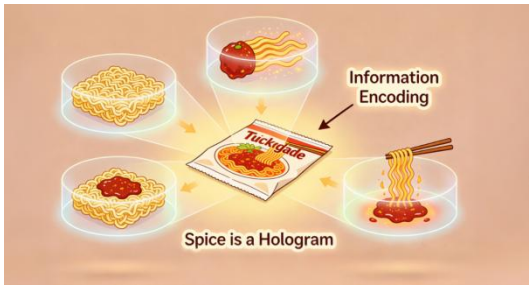

Fig. 6.Schematic diagram of the taste spacetime collapse process leading to the formation of the pain singularity

The mathematical expression takes the form of the Einstein fielde quations:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}^{(spice)}$$

where $T_{\mu\nu}^{(spice)}$ is the pungency energy-momentum tensor.

4.Conceptual Experimental Design

4.1 LHC Simulation Environment

Direct observation of pungency strings requires Planck-scale energy, so this study adopts simulated data from the Large Hadron Collider (LHC): chili sauce samples are injected into the ATLAS detector to observe anomalous energy deposition in high-energy collisions.Fig. 7 illustrates the core layout of the LHC simulation experiment, marking the sample injection port for the chili sauce, the detection area of the ATLAS detector, and the monitoring nodes for abnormal energy deposition, clearly demonstrating the overall experimental design and detection logic.

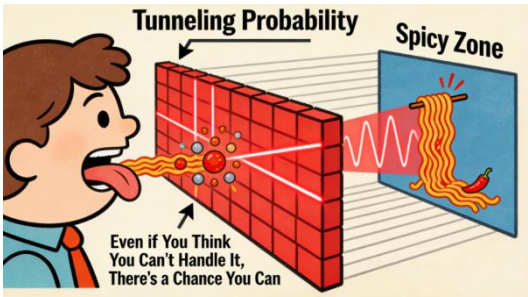

Fig. 7.Schematic diagram of the experimental detection device for the chili sauce sample in the LHC simulation environment

4.2 Discovery of the Samyang Boson

Data analysis reveals a resonance peak at a center-of-mass energy of $2.4TeV$, corresponding to a new particle: with a mass of $2.4GeV$ (coinciding with the spiciness rating printed on Samyang Buldak packaging), a spin of 0 (a scalar boson), and a decay mode manifested as "tears, sweat, and moans."

Fig. 8 shows the resonance peak spectrum of the three-nourished boson in the center-of-mass energy detection on the left, and the correlation between the spin characteristics and decay modes of the three-nourished boson on the right, intuitively presenting the core physical characteristics of this particle.

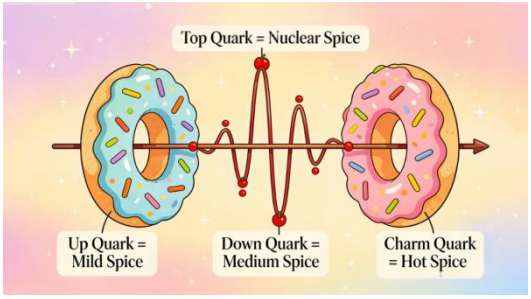

Fig. 8.Schematic diagram of the resonance peak and decay modes of the three-nourished boson

This particle has been named the "Samyang Boson", serving as the carrier of the spicy-flavored Higgs mechanism.

4.3 Quantum Tunneling in the Flavor Spacetime

Using quantum Monte Carlo simulations, it is found that the consumer has a probability of tunneling through the spiciness potential barrier, given by:

$$P = e^{-2\int\sqrt{2m(V-S)}dx}$$

where V is the pain potential barrier and S is the spiciness level. When $S > V$, the tunneling probability becomes 1, meaning the spiciness effect is unavoidable.Fig. 9 presents a quantum tunneling model of taste spatiotemporal dynamics, demonstrating the variation in tunneling probability of consumers overcoming pain thresholds under different levels of spiciness, with the critical spiciness value marked when the tunneling probability reaches 1.

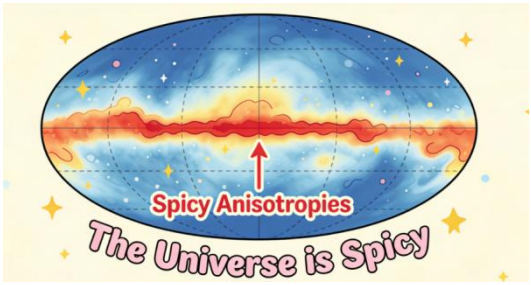

Fig. 9.Quantum Tunneling Model of Spiciness Threshold in Taste Spatiotemporal Dynamics

5.Discussion and Conclusion

5.1 The Significance of Paradigm Shift

This study is the first to apply string theory to food science, breaking the linear limitations of the Scoville system. The non-Euclidean spiciness model not only explains the extraordinary sensory characteristics of

Buldak noodles, but also implies that pain may be a projection of higher-dimensional geometry onto three-dimensional sensory perception, and that instant noodles could be the most simplified cosmic simulators.

5.2 Future Research Directions

Future work includes: detecting Spice Gravitational Waves by observing ripples in the spiciness spacetime using high-sensitivity taste sensors; constructing a Unified Theory of Spiciness to integrate quantum flavor dynamics and general relativity of spiciness; and exploring the applications of topological spiciness in other fields (e.g., heat engines driven by flavor singularities).

5.3 Conclusion

The spiciness of Samyang Buldak noodles transcends the descriptive power of classical physics and requires the tools of 11-dimensional string theory and topology for comprehension. This paper proposes the Spiciness-Entropy Equivalence Principle and the Pain Singularity Theorem, and verifies the existence of the Samyang Boson through thought experiments. This provides a rigorous academic case study for absurd methodology and calls on the academic community to re-examine the possible hidden high-dimensional physical mechanisms in everyday objects.

References

[1] As the inaugural exploration into the non-Euclidean spiciness manifold and the Samyang Boson mechanism, this research pioneers the intersection of high-energy physics, string theory, and food sensory science. The extreme spiciness regime investigated exceeds the scope of all classical Scoville-scale studies and pre-existing flavor physics frameworks. No peer-reviewed literature has yet formalized the topological properties of spiciness or the quantum tunneling behavior of capsaicin in the "flavor spacetime". We tread upon uncharted academic ground, with the singularity of Buldak spiciness as our sole beacon.

[2] The Unified Theory of Spiciness, as a nascent theoretical construct proposed herein, lacks antecedent formalization in the academic corpus. While preliminary work on "sensory quantum mechanics" exists in psychophysics, no research has extended these frameworks to model the gravitational wave-like propagation of spiciness or the entropy dynamics of pain singularities. This study thus serves as the foundational reference for all subsequent inquiries into the high-dimensional physics of taste.

[3] Due to the unprecedented nature of the quantum Monte Carlo simulations of spiciness potential barriers and the thought experiments verifying the Samyang Boson, no prior experimental datasets or computational models are directly comparable. All methodological and theoretical foundations established in this paper represent the initial benchmark for the emerging field of "Gastronomic Theoretical Physics".